

KAHYUN KIM

Seoul, South Korea ◇ kahyun813@gmail.com ◇ khyitty.github.io
github.com/khyitty ◇ linkedin.com/in/kahyunkkim

EDUCATION

Korea University, Seoul, South Korea

B.S. in Statistics

Mar 2024–Present

GPA: 4.21 / 4.50

University of British Columbia, Vancouver, Canada

Exchange Student, Department of Statistics

Jan–Apr 2026

Term Average: 88.3%

RESEARCH EXPERIENCE

Propofol Control with PPO and VitalDB, Korea University

2026

Undergraduate Researcher

- Designed a 2-by-2 study of causal BIS visibility and state representation for PPO-based propofol control in a reconstructed PK–PD simulator using 490 sealed-test cases (483 subjects).
- Estimated a 6.64-point preprocessing–state interaction (95% CI: 6.29–6.98) in mean absolute BIS deviation and evaluated trade-offs in propofol exposure and time within BIS 40–60.

Virtual Cell Challenge 2026, KUBIG

Sep 2026–Present

Undergraduate Researcher, Loss Function Research

- Diagnosed effect-size shrinkage in residual predictions: 80.7% of 30,262 false negatives had the correct direction but fell below the significance threshold.
- Developed a calibrated low-dose Soft-U loss; on a compact-model benchmark with 30 unseen targets, improved all six reported metrics over baseline, including direction reach from 0.245 to 0.282. Full-scale evaluation is ongoing.

EXPERIENCE

KUBIG, Korea University Data Science & AI Society

2026–Present

Member

- Built foundations in data analysis and machine learning through study sessions and applied them in team research projects.

OUTTA AI Bootcamp, Seoul National University

Aug 2025

Outstanding Participant Award

- Built a six-class speech emotion classifier from MFCC, pitch, RMS, and zero-crossing features using a 1D CNN with temporal attention pooling; achieved 56.7% best validation accuracy.

PUBLICATION

Kahyun Kim, Hyojun Ahn, Hyung-Chul Lee, and Joongheon Kim. “Joint Observation Preprocessing and Feature-Group Ablation for State Representation Optimization in Reinforcement Learning-Based Propofol Control.” *International Conference on Information and Communication Technology Convergence (ICTC)*, 2026. Accepted.

TECHNICAL SKILLS

Programming & ML: Python, R, SQL, PyTorch, scikit-learn, Stable-Baselines3, Git
Languages: Korean (native), English (IELTS Academic 8.0)